

Department of Electrical and Electronic Engineering

Islamic University of Technology (IUT)
A subsidiary organ of the OIC

Project Manual

EEE 4308: Digital Electronics Lab

Name of the Project:

SAP-1 Architecture-based 8-bit Computer Design and Implementation

Project Description

Objective:

The objective of the project is to implement a 8-bit computer based on the architecture of SAP-1 (Simple-As-Possible) using combinational and sequential circuits.

Project Specification:

The computer must have the following features and functionalities-

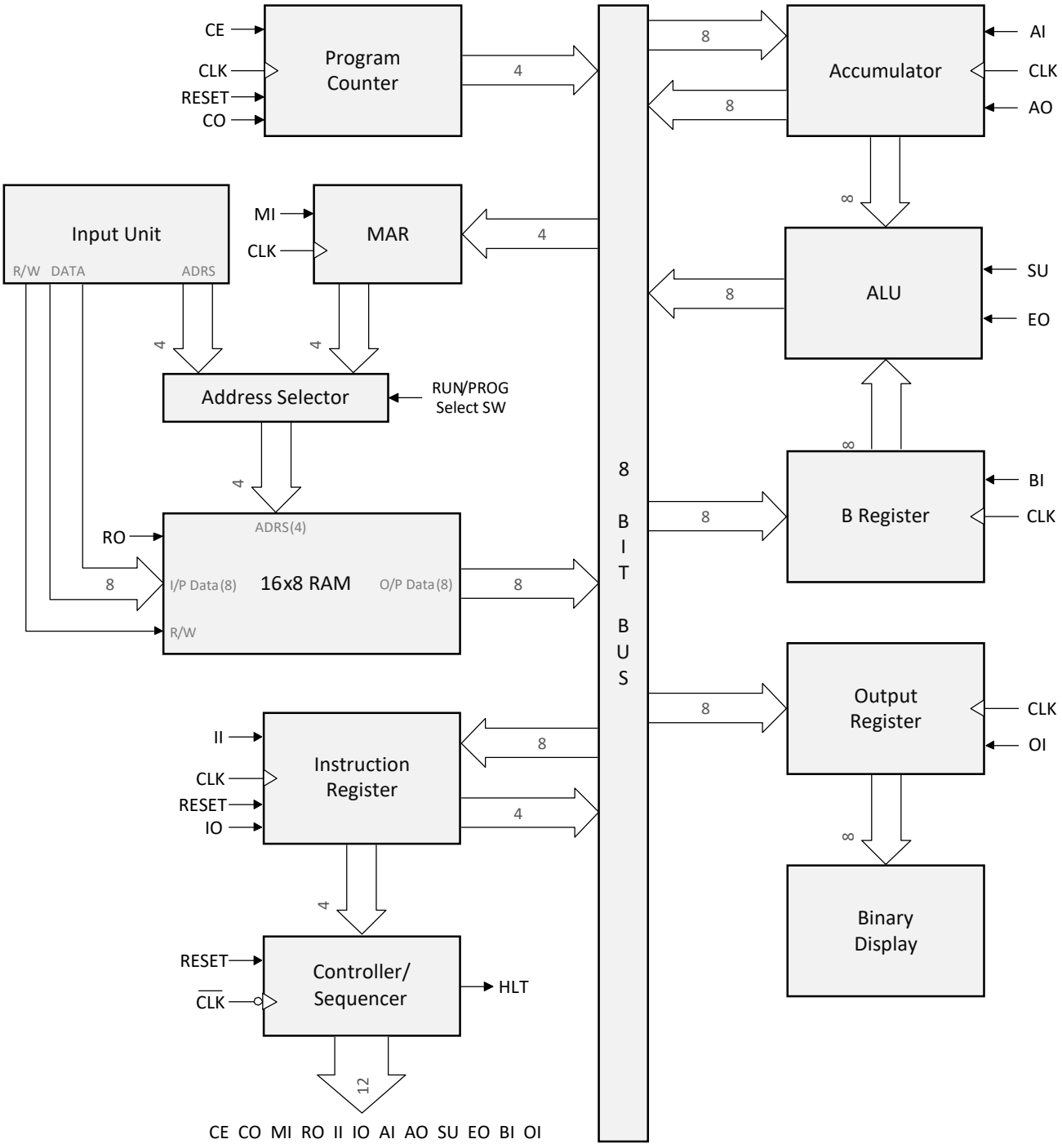
1. A single 8-bit bus for address and data transfer.
2. 16x8 bit random access memory (RAM) for storing program and data.
3. A program counter for containing the address of the next instruction (should be capable of counting from 0000 to 1111).
4. Input unit for writing the program into RAM before running the computer.
5. Memory address register (MAR) for accessing RAM data.
6. An arithmetic and logic unit (ALU) capable of 8-bit addition and subtraction. Two registers Accumulator (register A) and register B (both 8-bit) are also needed for storing data for ALU operation.
7. A control unit for controlling all other blocks (registers, memory, ALU etc.)
8. Output unit capable of showing the output in binary form.
9. Must have the ability to perform following instruction set-

Operation	Description
Load	Load data from memory to the accumulator
Add	Add data from memory to value in the accumulator
Subtract	Subtract data from memory from value in the accumulator
Output	Load data from accumulator to the output register
Halt	Stop processing

10. Main clock signal driving the computer should have two modes of operation- (i) Auto and (ii) Manual.
11. The computer must have all the blocks shown in the project diagram (Fig. 1). Specifications of each block are described later.

A complete diagram of the project is shown on the next page.

Project Diagram:



Submodule/Block Specification:

This section discusses the specifications/requirements for each block/submodule shown in the project diagram (Fig. 1) -

8-bit Bus:

1. Bus must have 8 lines for address and data transfer.
2. It should be designed in such a way, so that at a single time instance only one module will send data to the bus, and only one module will receive that data.
3. Two modules can never send data to the bus at the same time (bus contention). When a module is sending data to the bus, all other modules' outputs must be disconnected.

Program Counter:

Program counter (PC) is a 4-bit counter which can count from 0000 up to 1111. Program counter's job is to store and send out the memory address of the next instruction to be fetched and executed. It will increment its value after each instruction cycle (completion of an instruction). Functions of its I/O pins are as follows-

Pin	Type	Function
CE	In	Program Counter Enable: Increments the program counter on the next clock-cycle.
CO	In	Program Counter Out: Puts the current program counter value on the bus. Output is disconnected when CO is low.
RESET	In	Resets the program counter to 0000
CLK	In	Clock pulse input.
OUT	Out	4-bit counter output.

Input Unit and Memory Address Register (MAR):

Memory address register (MAR) is a 4-bit register. MAR stores the 4-bit address of data or instruction which are placed in RAM. When the computer is in the RUN state, the 4-bit address is obtained via the bus from the Program Counter and then stored. This stored address is sent to the RAM where data or instructions are read from. Functions of its I/O pins are as follows-

Pin	Type	Function
MI	In	Memory Address In: Stores the current values of the bus into the Memory address register (MAR).
CLK	In	Clock pulse input.
OUT	Out	4-bit address data.

Input unit is used when computer is needed to be programmed by writing data into the RAM (PROG state). There should be a selector to choose between MAR output (4-bit) and Input address output (4-bit) depending on the state (RUN/PROG). The output of selector should be connected to address pins of RAM. The following is a description of switches that input unit should contain-

Switches	Description
Input data switches	8 switches (D7-D0) for writing 8-bit data to specific address.
Address switches	4 switches (A3-A0) for accessing a particular address of RAM for writing.
R/W switch	For switching between read and write mode.

Random Access Memory (RAM):

RAM size should be 16 x 8 bit (16 memory locations each storing 8-bits of data). The RAM will be programmed by means of the address switches and data switches of input unit. During a computer run, the RAM will receive its 4-bit address from the MAR.

Pin (Bit)	Type	Function
ADRS (4)	In	4-bit address input for accessing RAM locations.
I/P Data (8)	In	In PROG state, used for writing 8-bit data to addressed location.
R/W	In	Read or write mode.
CS	In	Chip select pin.
RO	In	RAM Out: Put the currently selected RAM byte onto the bus. Output is disconnected when RO is low.
O/P Data (8)	Out	8-bit output data.

Accumulator, B Register and Output Register:

All of these registers will be of 8-bit size. Accumulator (also called as A register) should have the ability to output intermediate results after each ALU operation. Accumulator and B register will be used to store 8-bit data as input to ALU input. The purpose of the output register is to store result to display when requested. Controller/sequencer unit sends control signals to these registers for controlling input/output enable. Description of these control bits-

Pin	Name	Function
AI	Accumulator In	Stores the current values of the bus into Accumulator.
AO	Accumulator Out	Accumulator sends its stored content to the bus. Output of accumulator is disconnected when AO is low.
BI	B Register In	Stores the current values of the bus into B Register.
OI	Output Register In	Stores the current values of the bus into Output Register.

Arithmetic and Logic Unit (ALU):

ALU should be capable of 8-bit addition and subtraction. For addition and subtraction, 2's complement approach should be used. Description of two control bits of ALU-

Pin	Name	Function
SU	Addition/Subtraction Select	When SU=0, addition operation and when SU=1, subtraction operation will be executed.

EO	ALU Out	Puts the ALU output onto the bus. Output of ALU should be disconnected when RO is low.

Instruction Register:

The function of instruction register is to receive and store the 8-bit instruction placed on the bus from the RAM. The contents of the instruction register are then split into two "4-bit segments". The "upper 4-bit" goes into the controller-sequencer while the "lower 4-bit" should be sent to the bus. A reset pin is required for resetting the instruction register when the computer starts. Control bits of instruction register are-

Pin	Name	Function
II	Instruction Register In	Writes the current values of the bus into the instruction register.
IO	Instruction Register Out	Sends the stored data (only lower nibble) onto the bus. Output pins are disconnected when AO is low.

Controller-Sequencer:

Controller-sequencer generates necessary control signals for each block so that actions occur in a desired sequence. 12 control bits come out of the controller-sequencer block. These control bits determine how all other blocks will react to the next positive CLK edge. So, controller-sequencer must be designed in such a way so that correct control bits are already available to all the blocks before the next positive CLK edge. 12 control bits are –

CON = CE CO MI RO II IO AI AO SU EO BI OI

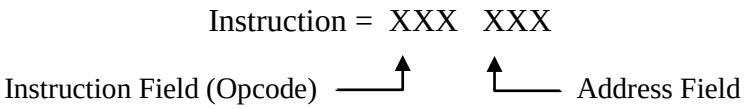
Functions of all 12 control bits are already described. Besides these 12 signals, controller-sequencer also generates a HLT (halt) signal, which can halt the process of computer by stopping the main clock.

Main Clock Signal:

Clock signal should be generated using 555 timer IC. The clock module needs to have two modes of operation available- (i) Auto and (ii) Manual. In manual mode, pressing a switch/push button will advance each clock cycle.

Specification of Instruction Set:

Each instruction of this 8-bit SAP-1 computer should be of 8-bits length. The upper 4-bits of this instruction (bits 7-4) is called "opcode", and it defines the instruction. The lower 4-bits (bits 3-0) can be used to pass parameters to the instruction (for example a RAM Address).



The computer must be able to perform the following five instructions-

Instruction	Mnemonic	Opcode	Description
Load Accumulator	LDA	0000	Load the value stored in a particular RAM address which is specified by the address field of the instruction, then store it in the accumulator.
Add	ADD	0001	Add the value stored in a particular RAM address to the value in the accumulator. RAM address is given by address field of instruction.
Subtract	SUB	0010	Subtract value stored in a particular RAM address from the value in the accumulator. RAM address is given by address field of instruction.
Output	OUT	0011	Load data from the accumulator and send that to the output register.
Halt	HLT	1111	Stop processing.

Instruction Cycle:

After each instruction is executed, the program counter points to the next instruction stored in RAM. But each instruction consists of several smaller instructions called **microinstruction**. The time required by the CPU to execute one single instruction is called **instruction cycle**. For SAP-1, length of this instruction cycle is 6-time states (T states), where **T state** is the time required to complete one microinstruction.

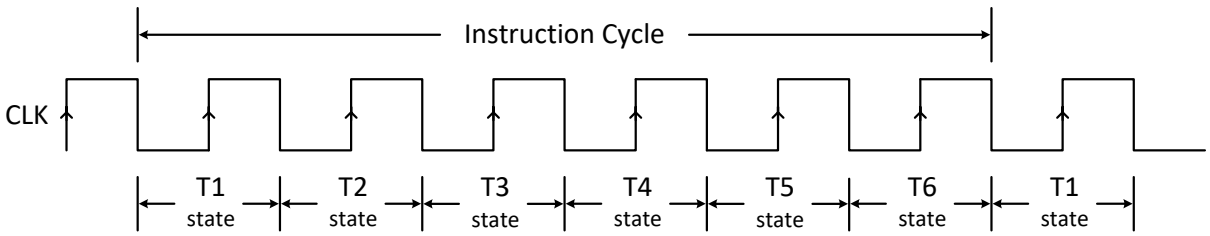


Fig. 2: T1-T6 states in SAP-1 computer.

At the beginning of a T state, controller-sequencer modifies the value of 12 control bits depending on the microinstruction. During that T state, all control bits are kept unchanged. Then, at the start of next T state, control bits are again modified based on the requirement of next microinstruction.

One important thing to observe, each T state starts at the negative edge of the clock pulse (fig. 2), where all the submodules/blocks of the computer are triggered at the positive edge of clock pulse. This is required, so that correct control bits are already available to all the submodules/blocks before the submodules trigger.

Instruction cycle can be further divided into two parts-

i. Fetch cycle (First 3 T states: T1, T2, T3)
ii. Execution cycle (Last 3 T states: T4, T5, T6)

Fetch Cycle:

Fetch cycle (T1, T2, T3) is responsible for fetching the instruction from memory. Fetch cycle is exactly the same for all instructions. So, register transfers (or microinstructions) during T1-T3 states will be always same for SAP-1 for any instruction.

T State	Name	Activated Control Bits	Function
T1	Address State	CO MI	Address stored in the program counter (PC) is transferred to the memory address register (MAR).
T2	Increment State	CE	Program counter (PC) is incremented.
T3	Memory State	RO II	Instruction stored at the addressed location of RAM is transferred to instruction register.

Execution Cycle:

During execution cycle (T4, T5, T6) fetched instruction gets executed. Microinstructions during these three states can vary depending upon the instruction that was fetched. Execution cycle for each instruction (LDA, ADD, SUB, OUT, HLT) are as follows-

LDA Routine

T State	Activated Control Bits	Function
T4	IO MI	Upper nibble of instruction register output (IR) goes to controller-sequencer. Lower nibble is loaded into MAR.
T5	AI RO	Addressed data of RAM gets loaded into the accumulator.
T6		No operation.

ADD Routine

T State	Activated Control Bits	Function
T4	IO MI	Upper nibble of instruction register output (IR) goes to controller-sequencer. Lower nibble is loaded into MAR.
T5	RO BI	Addressed data of RAM gets loaded into B register.
T6	EO AI	ALU output is loaded into accumulator.

SUB Routine

T State	Activated Control Bits	Function
T4	IO MI	Upper nibble of instruction register output (IR) goes to controller-sequencer. Lower nibble is loaded into MAR.
T5	RO BI	Addressed data of RAM gets loaded into B register.
T6	EO AI SU	SU bit sets up the ALU to do subtraction. Then subtraction output of ALU gets loaded into the accumulator.

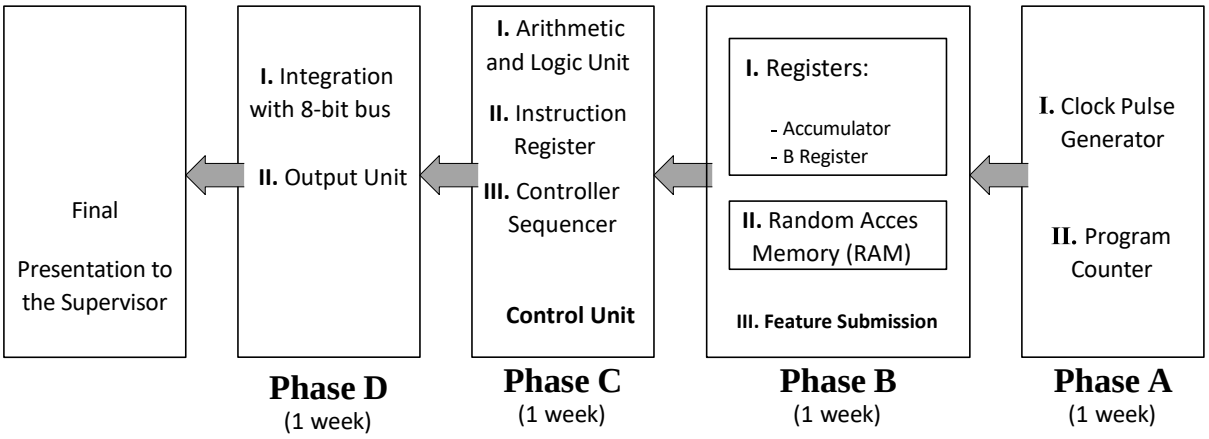
OUT Routine

T State	Activated Control Bits	Function
T4	AO OI	Content of accumulator is stored in output register.
T5		No operation.
T6		No operation.

HLT Routine

HLT does not require any control routine because no registers are involved in execution of halt instruction. After HLT instruction is fetched, controller-sequencer activates the HLT bit, which stops the main clock.

Evaluation Process



Rubrics for Project:

Criteria	Scoring Level				Maximum percentage out of the total marks for the project
	Excellent	**Good**	**Satisfactory**	**Not Satisfactory**	
	100% of total percentage allotted	80% of total percentage allotted	60% of total percentage allotted	= or < 40% of total percentage allotted	
Phase A	Submodules of Phase A are perfectly designed and working with best performance	Submodules of Phase A are designed adequately but needs further modification	Submodules Phase A are designed moderately and working with errors	Submodules of Phase A are poorly designed and not working	**20%**
Phase B	Submodules of Phase B are perfectly designed and working with best performance	Submodules of Phase B are designed adequately but needs further modification	Submodules Phase B are designed moderately and working with errors	Submodules of Phase B are poorly designed and not working	**20%**
Phase C	Submodules of Phase C are perfectly designed and working with best performance	Submodules of Phase C are designed adequately but needs further modification	Submodules Phase C are designed moderately and working with errors	Submodules of Phase C are poorly designed and not working	**25%**
Phase D	Submodules of Phase D are perfectly designed and working with best performance	Submodules of Phase D are designed adequately but needs further modification	Submodules Phase D are designed moderately and working with errors	Submodules of Phase D are poorly designed and not working	**35%**

- The marks allocated on the above rubrics are as percentage of the full marks allowed for the **Project Completion** only. Marks distribution of EEE 4308 course is provided below:

Assessment Method	Marks (as %)
Lab Performance	15%
Lab Report	10%
Open Ended Lab	15%
Project Completion	**50%**
Project Presentation	5%
Project Report	5%

- In each of the criteria mentioned in the rubrics, **40%** of that section's marks will be for **software simulation** and **60%** of the mark will be for **hardware implementation**. For example, if 15% of the total project mark is allocated for Phase B, then 6% mark is for software simulation and 9% mark is for hardware implementation in Phase B.

- If a group fails to complete a Design Phase within its deadline and submits after the deadline, they will be **penalized by one scoring level** for each one-week delay .

  For example, if a group submits Design Phase B after its given deadline and their submitted design is working perfectly, still they will not get "Excellent" scoring level (100%) for Design Phase B. They will get "Good" scoring level (80%), if their delay is less than a week. If they miss the deadline by more than one week, they will receive a "Satisfactory" score (60%). If the delay is more than two weeks, they will get "Not Satisfactory" score (= or < 40%).

Mapping of Rubrics and Program Outcomes (POs):

Criteria	Program Outcomes (POs)											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Phase A		√	√		√				√			
Phase B		√	√		√				√			
Phase C		√	√		√				√			
Phase D		√	√		√				√		√	

Mapping of Rubrics and Program Outcomes (POs) with Marks Distribution:

Criteria	Marks (as %)	Mark distributions (as %) on POs				
		PO2	PO3	PO5	PO9	PO11
Phase A	20%	5%	5%	5%	5%	0%
Phase B	20%	5%	5%	5%	5%	0%
Phase C	25%	6.25%	6.25%	6.25%	6.25%	0%
Phase D	35%	7.25%	7.25%	7.25%	7.25%	6%
Total	**100%**	**23.5%**	**23.5%**	**23.5%**	**23.5%**	**6%**

For continuous evaluation and supervision of each group, one course teacher will be designated. For each design phase, all groups must **demonstrate their project progress** and **submit a progress report** to their respective supervisors. At the same time, they can consult with their supervisor whenever they face any problem regarding the project. The list of project groups and their supervisor is given below:

Group Name	Supervisor
Boolean Brigade	Mr. Mohd. Abu Bakar Siddique
Tartans	
Gray Coders	
Cassette	
Sir It worked yesterday	Mr. Aseer Imad Keats
Silicon 6	
Six Segments	
Team Wrong Output	
E-builders	Ms. Sumaiya Bashar
The Backbenchers	
Logic 7	
Bitsolver	
AC/DC	Mr. Ahmed Jawad Rashid
Team JFK	
Team Boole	
Corona Effect	
Team NOR-mal	Ms. Fatema Tasnim Oyshi
Hexcorp	
The Peaky Blinders	
Boolean Squad	
-15 Marks	Mr. Md. Rahib-Bin-Hossain
AUSTAD	
Byte_Masters	
Inversion	

Suggested List of Components:

- 555 timer IC must be used to generate clock pulse.
- The following is a list of suggested ICs that may be used in the project:

Functional Name	Type
Basic Logic Gate	AND, OR, XOR, NAND, NOR, XNOR, NOT (2 input, 3 input, 4 input, 5 input, etc)
Decoder	2 to 4, 3 to 8, 4 to 10, 4 to 16, BCD to Seven segment
Encoder	4 to 2, 8 to 3, 10 to 4, 16 to 4
Multiplexer	2 to 1, 4 to 1, 8 to 1
Adder	Binary
Flip-flop	J-K, D, T, Master-Slave
Register	4-bit, 8-bit
Other ICs	Tri-state Buffer, Random Access Memory

However, if you need to use any IC other than the ICs mentioned above, you are advised to take prior permission from your course teachers.

Feature Submission:

As part of the project assessment, marks will be awarded for any innovative or effective optimization incorporated into the system design. This feature submission encourages students to go beyond the basic implementation and demonstrate critical thinking, efficiency, and creativity in improving the project's performance, complexity, or usability.

Optimization may be applied to any relevant part of the system. Students are required to submit their proposed feature or optimization to their respective project supervisor **within Phase B** of the project timeline. The submission should include a clear description of the optimization, its implementation approach, and an explanation of how it enhances the overall system.

Report Outline:

- Title Page: Course code and name, Project name, Group Name, Group members' name and student ID
- Main Body: Objective and brief description of the Project
  For Each Phase: Block Diagram, algorithm, Truth Table, Boolean expression, Circuit Diagram, Equipment list, PCB design diagram, pictures of hardware implemented Phase.
  Complete Project: Block Diagram, Circuit Diagram, PCB design diagram, pictures of hardware implemented Project.
- Problems Faced
- Discussion: Outcomes, Limitations, Future development

Project Implementation and Submission Timeline:

Task	Submission Deadline*
Project Introduction	January 16, 2026
Demonstration and Progress Report: Phase A	January 30, 2026
Demonstration and Progress Report: Phase B	February 6, 2026
Demonstration and Progress Report: Phase C	February 13, 2026
Demonstration and Progress Report: Phase D	February 20, 2026
Complete Project Submission	**February 24, 2026**
Project Demonstration/Presentation	February 27, 2026
Project Report Submission	March 2, 2026

*****on or before

The project groups are highly encouraged to submit the complete project before the deadline which will provide extra advantage in getting better grades.

A note on PLAGIARISM

Plagiarism is the misrepresentation of the work of another as your own. It is academic theft; it is a serious infraction of the University Honor Code. Instances of plagiarism or any other cheating will be reported to the Department and will at the very least result in failure of this course.